Why Contact

Paint a contact cut between poly and metal1

The idea

- GOLDEN: contact_count=1 (ca)

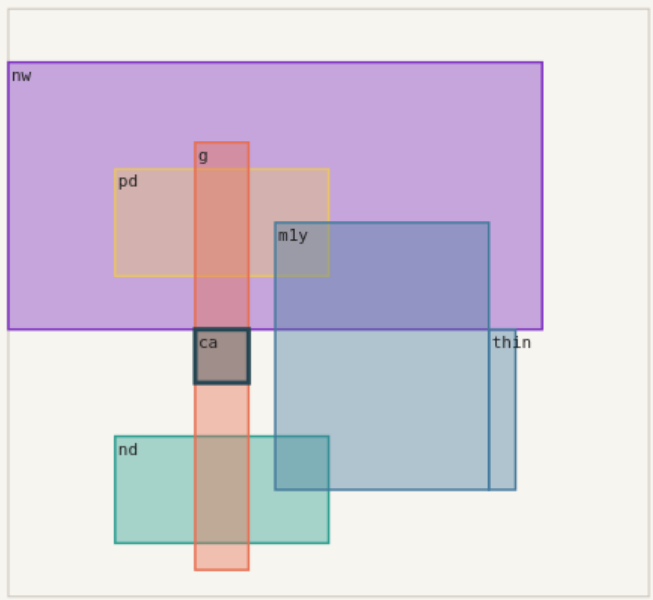
Contact ca

CONTACT PAINT

Contact ca

Step 1 / 5 · ca · module02-01-contact-paint

1



Explanation

One contact between poly and metal1.

- ca

```
1
```

Contact ca

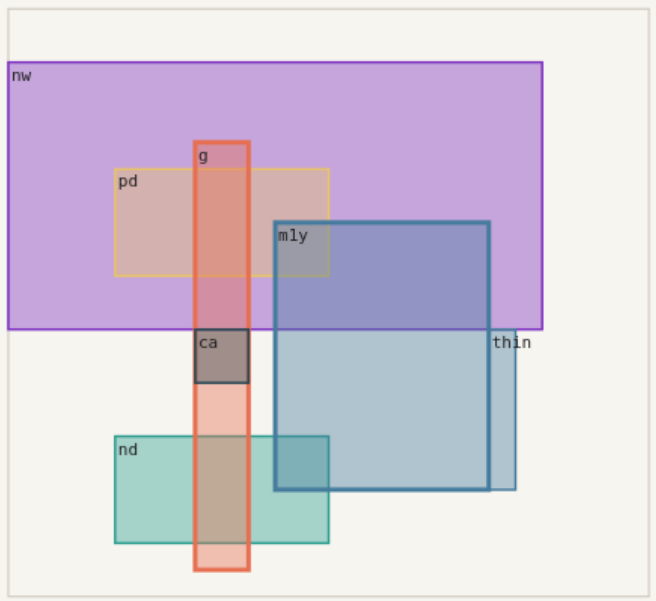
Between layers

CONTACT PAINT

Between layers

Step 2 / 5 · between · module02-01-contact-paint

bridge



Explanation

Contact bridges poly (g) and metal1 (m1y).

- g
- m1y

```
bridge
```

Between layers

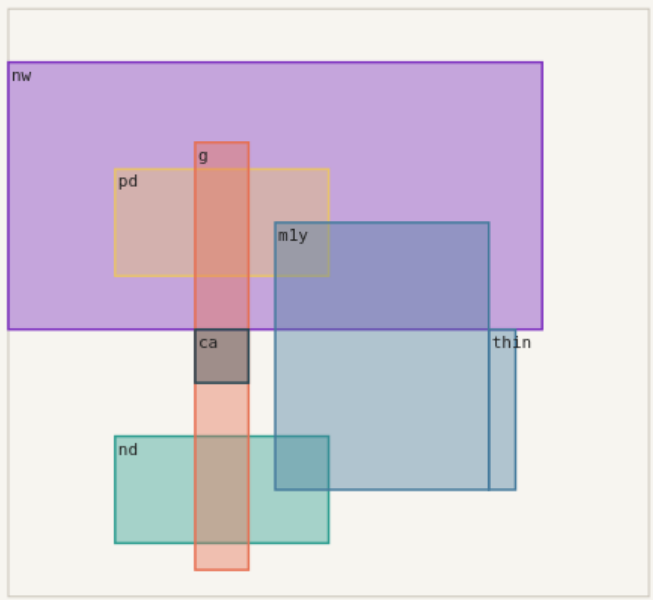
Contact paint

CONTACT PAINT

Contact paint

Step 3 / 5 · paint · module02-01-contact-paint

paint



Explanation

Magic contact command paints the cut.

- contact

```
paint
```

Contact paint

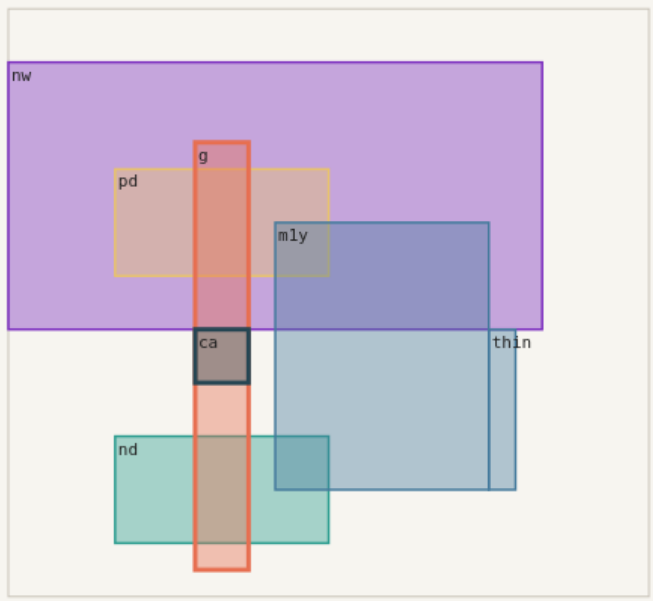
Highlight

CONTACT PAINT

Highlight

Step 4 / 5 · highlight · module02-01-contact-paint

3



Explanation

Contact + gate + m1y highlighted.

- ca
- g

```
3
```

Highlight

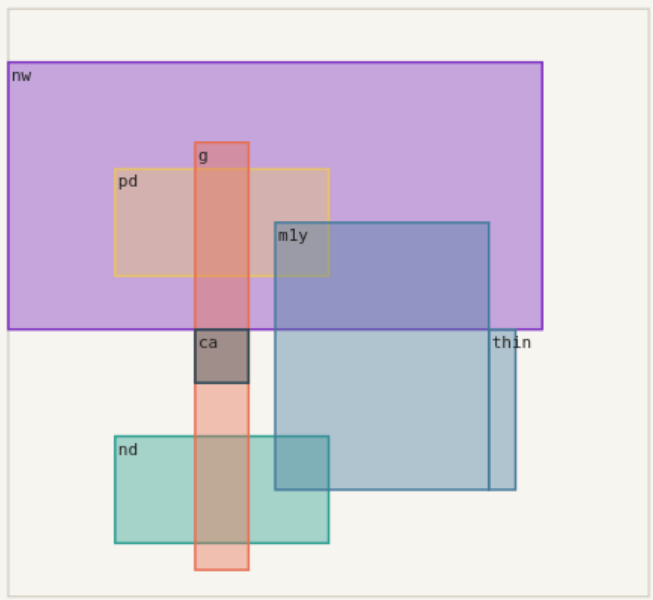
Golden

CONTACT PAINT

Golden

Step 5 / 5 · golden · module02-01-contact-paint

1



Explanation

GOLDEN: contact_count=1.

- GOLDEN

```
1
```

Golden

Browser lab track

- Open **contact-paint**
- Run the stage buttons and Check the goldens

Implement track

- From common/, run solvers.py and confirm MAGIC_PARTA_OK

Pitfalls

- Mixing KLayout viewer shortcuts with Magic paint commands
- Forgetting min_width when reading DRC errors

Try these

- From the common folder, run the Magic map self-test

Try these — try these

```
cd courses/learn_magic/common
python3 solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
contact_count=1 (ca).
ok if goldens match
```

Next

- Complete the quiz for this part
- Continue the Magic Part A map, then the real CLI track